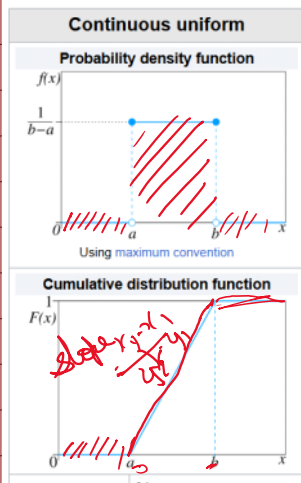




- i) Continuous uniform distribution ^{b+1} $\rightarrow$ P.d.f
 ii) Discrete uniform distribution $\rightarrow$ p.m.f

i) Continuous uniform distribution

Notation: $U(a, b)$

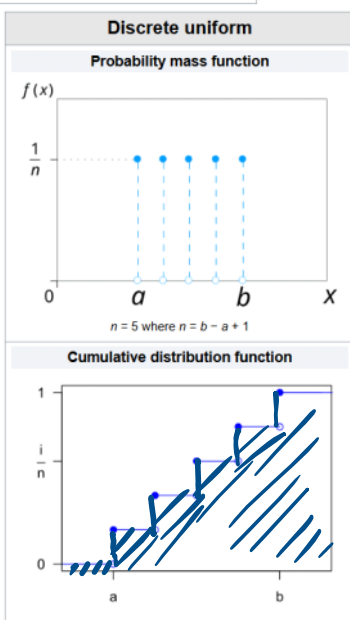
$$-\infty < a < b < \infty$$

$$pdf = \begin{cases} \frac{1}{b-a} & x \in (a, b) \\ 0 & \text{otherwise} \end{cases} \quad 0,1$$

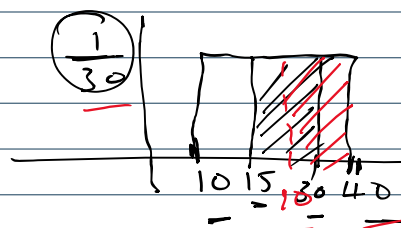
ii) Cumulative distribution function (CDF)

$$cdf = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & x \in (a, b) \\ 1 & \text{for } x > b \end{cases}$$

Eg: - No. of Samosas sold at a stall is uniformly distributed with a maximum of 40 Samosas and min of 10



i) Probability of daily sale to fall between 15 & 30



$$\frac{1}{40-10} = \frac{1}{30}$$

$$\begin{aligned} p(15 \leq x \leq 30) &= (x_2 - x_1) \frac{1}{b-a} \\ &= (30 - 15) \times \frac{1}{30} \\ &= 15 \times \frac{1}{30} = \frac{1}{2} \Rightarrow 50\% \end{aligned}$$

$$p(x > 20)$$

$$20 \leq x \leq 40 = 40 - 20 \times \frac{1}{30} \Rightarrow 20 \times \frac{1}{30} = 0.6$$

$$pX(x > 20)$$

$$20 \leq x \leq 40 = 40 - 20 \times \frac{1}{20} \Rightarrow 20 \times \frac{1}{20} = 1 \Rightarrow 0.6 \Rightarrow 60\%$$

1) Discrete uniform distribution

① Discrete values

② pmf $\rightarrow$ Probability mass function

Ex: - Rolling a die $\{1, 2, 3, 4, 5, 6\}$

①

②

$$\frac{1}{6} - \frac{1}{6}$$

$$\frac{1}{n} \Rightarrow n = b - a + 1 \text{ where } a, b$$

$$b \geq a$$

$$pmf = \frac{1}{n}$$

